

AI Usage Note

AlgoChowk — Quant Engineer Intern Challenge | Sumit Yadav | 22 Sep 2026 | github.com/sumityadav026/Algochowk-Nifty-Research

Tools used

Claude (Anthropic), used throughout via conversational chat — no autonomous code-writing tool (e.g. Cursor/Copilot) was used.

How I used it

Claude was used in three ways: (1) to scaffold the codebase — data loader, event-detection engine, statistics module, and backtest — against a research design I specified up front (event definition, entry/exit rule, holding period, cost assumptions, chronological train/test split); (2) to explain statistical concepts I hadn't applied before (bootstrap resampling, the Holm correction for multiple testing, look-ahead bias) until I could connect them to ideas I already knew from machine learning — look-ahead bias as a form of data leakage, bootstrap as a resampling method similar in spirit to cross-validation, and the chronological split as the time-series analogue of a train/test split; and (3) to help build an independent cross-check script that re-derives the core statistics with a separate, plain-loop implementation, so the main results are not resting on a single code path.

My own decisions

- Event threshold: I fixed -2.0% as the single main rule before running any experiment. It is a round, defensible number — roughly twice NIFTY's typical daily volatility — large enough to represent a genuine outlier without being so rare that too few events remain for a meaningful sample. -1.5% and -3.0% were used only afterward, as robustness checks, never as a search for a better-looking number.
- Entry timing: I required entry to occur at the next trading day's open rather than the event day's own close, since the -2% return is only known once that close has already happened; buying at that close would use information not yet available at decision time.
- Verification before trust: rather than accept the first run's output, I asked for the event detection, statistics, and backtest to be independently re-derived with a separate implementation (`tests/cross_check.py`), and treated the results as reliable only once the two approaches matched exactly.
- Data integrity: I manually verified three of the largest single-day falls in the dataset (2020-03-23, 2008-10-24, 2008-01-21) against an independent public source before trusting the downloaded data, since an error in even one of ~ 95 events could meaningfully distort a sample this size.

Suggestions I questioned / verified rather than accepted

- I did not accept the entry-timing logic on explanation alone — I asked for and reviewed the specific line of code and its accompanying unit test confirming the strategy buys at the next day's open, not the event day's close, before treating the look-ahead-bias fix as real rather than assumed.
- When the robustness grid showed only 21 events for a “worst 5% of days” event rule versus 95 for the flat -2% rule, I treated the mismatch as something to explain, not ignore — the diagnostic showed the two rules had converged to nearly the same threshold (-2.07%) by the end of 2018, which resolved the discrepancy rather than hiding it.
- When excluding the 2008–09 and 2020 crisis clusters flipped the sign of the excess return from negative to positive, I made sure this was reported as still statistically insignificant (confidence interval includes zero), rather than presented as a hidden positive result — a changed sign is not the same as a confirmed effect.

What I learned

The concept I now understand far better than before is why a result has to beat a baseline, not just be positive, to mean anything. NIFTY drifts upward over time regardless of what happens on any single day, so a positive average return after a crash could simply reflect that normal upward drift rather than any special “recovery.” Comparing event returns only to zero would have overstated the case for the hypothesis; comparing them to a matched non-event baseline over the same period is what actually tests the claim being made — and in this project, once that comparison was made, the apparent “recovery” disappeared.

I also learned that excluding an extreme period can change the sign of a result without making that result meaningful: the sign flipped when the 2008/2020 crisis clusters were removed, but the confidence interval still contained zero. That distinction — between a changed sign and a confirmed effect — is now something I check for by default rather than something I would have previously overlooked.